

CSA15 Cloud Computing and Big Data Analytics

Local IaaS Lab: Type-2 Hypervisor Installation and First Linux Virtual Machine

Course Code	CSA15	Course Title	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-01	Submission	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Target Learner	Beginner CSE student	Estimated Duration	3 to 4 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation + selected Linux ISO	Cloud Link	Models the local foundation of IaaS before public-cloud VM provisioning
Guided Project Pattern Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom.			

Learning Outcomes

- Install either Oracle VirtualBox or VMware Workstation as a type-2 hypervisor.
- Select a guest operating system according to download size, laptop capacity and lab time.
- Create and configure a VM with CPU, RAM, virtual disk, storage controller and NAT network.
- Boot or install the selected guest OS and verify CPU, memory, storage and network evidence.
- Submit an individual evidence portfolio through GitHub and Google Classroom.

Before You Start

- Windows/macOS/Linux laptop with virtualization enabled in BIOS/UEFI or system settings.
- Minimum 8 GB host RAM recommended; keep at least 25 GB free disk space when using Ubuntu and at least 8 GB when using Tiny Core/Alpine.
- Administrator permission to install the hypervisor and its network drivers.
- One selected hypervisor and one selected guest OS image from the tables below.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A host OS is the operating system already running on the laptop. A guest OS is the operating system running inside the VM.
- A type-2 hypervisor runs above the host OS, so it is suitable for student laptops and lab demonstrations.
- An ISO file behaves like a bootable installation DVD attached to the VM.
- NAT networking is preferred for this beginner lab because the guest VM can access the internet through the host without being directly exposed to the local network.
- The value 1 CPU, 2 GB RAM and 15 GB disk is the standard lab configuration; smaller OS images may safely use lower RAM/disk values if the student records the reason.

Lab Exercise Coverage

Lab Exercise	How This Guided Project Covers It
Experiment 7	Install a type-2 hypervisor, create and configure a VM image with a selected guest operating system.
Experiment 8	Create a VM with 1 CPU, 2 GB RAM and 15 GB storage disk, or record a justified lightweight-OS alternative.

Hypervisor Choice

Tool	When to Choose It	Student Evidence
Oracle VirtualBox	Choose when the laptop supports VirtualBox well, the student wants a free cross-platform tool, or VMware is not available.	VirtualBox version, New VM window, Settings screenshots and running guest OS proof.
VMware Workstation	Choose when VMware installer is available and the student wants to practice the lab-manual VMware workflow.	Installer file/version, VMware home screen, VM settings screenshots and running guest OS proof.

Guest OS Selection Guide

Choose one guest operating system based on laptop capacity, internet bandwidth and lab time. Record the selected image name, approximate size and reason for selection before creating the VM.

Option	Approx. Size	Best Use	Official Download	Suggested VM Setting
Tiny Core Linux Core	About 17 MB	Fastest boot and very low-bandwidth VM proof.	https://tinycorelinux.net/downloads.html	1 CPU, 256 to 512 MB RAM, 2 to 4 GB disk
Tiny Core Linux CorePlus	About 250 MB	Graphical desktop practice with small download size.	https://tinycorelinux.net/downloads.html	1 CPU, 512 MB RAM, 4 GB disk
Alpine Linux Standard	About 200 to 300 MB	Lightweight server-style Linux and package practice.	https://alpinelinux.org/downloads/	1 CPU, 512 MB to 1 GB RAM, 4 to 8 GB disk
Debian netinst	About 700 to 800 MB	Minimal standard Linux installation with optional packages.	https://www.debian.org/distrib/netinst	1 CPU, 1 to 2 GB RAM, 8 to 15 GB disk
Ubuntu Server LTS	About 3 to 4 GB	Full Linux server practice when bandwidth is available.	https://ubuntu.com/download/server	1 CPU, 2 GB RAM, 15 GB disk
Ubuntu Desktop LTS	About 6 GB	Full desktop Linux practice when bandwidth and storage are available.	https://ubuntu.com/download/desktop	1 CPU, 2 to 4 GB RAM, 20 GB disk

Safety Rule

Do not allocate all host resources to the VM. Use NAT networking unless instructed otherwise. Never select a physical host disk for guest installation; the installation must happen only inside the virtual disk.

Architecture View

Laptop Hardware	Host OS	Selected Type-2 Hypervisor	Virtual Hardware	Guest OS ISO	Evidence
-----------------	---------	----------------------------	------------------	--------------	----------

The laptop remains the physical resource provider. The selected hypervisor creates virtual CPU, RAM, disk and network devices. The selected Linux image becomes the guest operating system.

Guided Build

Stage 1 - Confirm Host Readiness

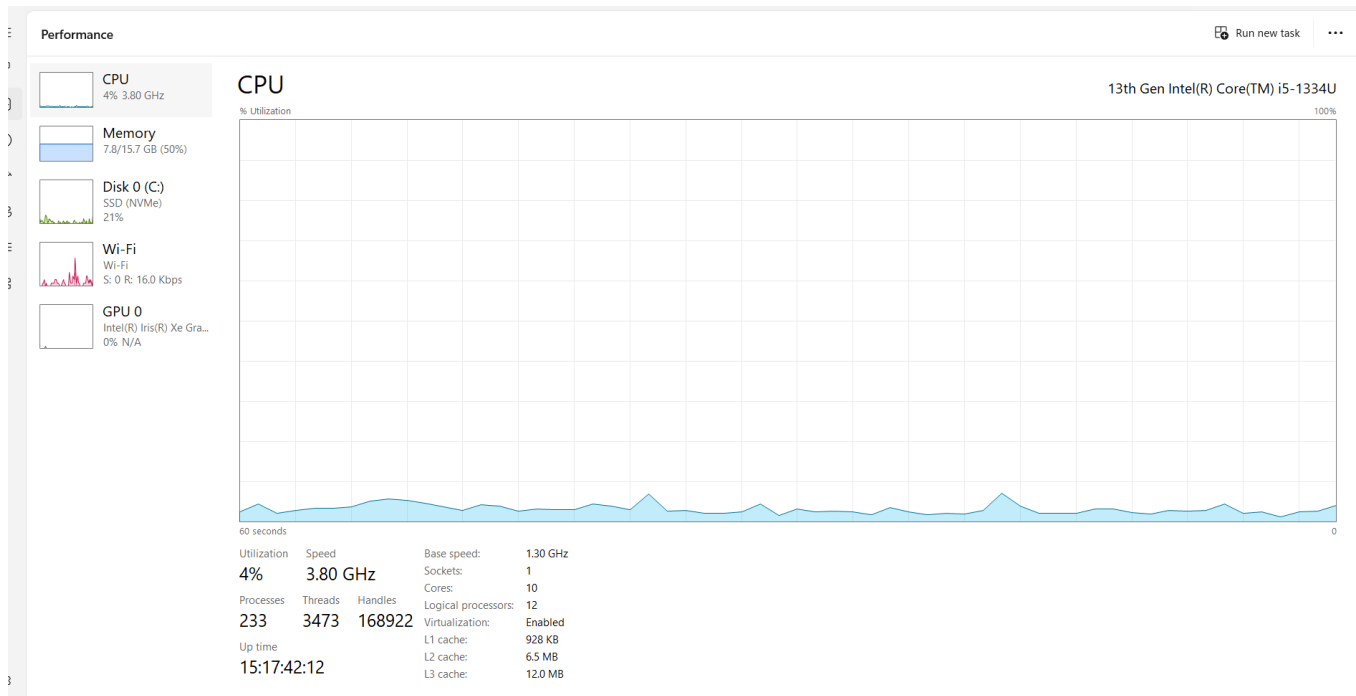
Learn: Virtualization support must be available before a VM can run reliably.

- Open Task Manager on Windows and check Performance -> CPU -> Virtualization. On macOS/Linux, use system information or terminal/system settings as available.
- Record host operating system, processor type, installed RAM and free disk space.
- Create a folder named CSA15_VM01_RegisterNumber on the host system.
- If virtualization is disabled, record the issue and enable it through BIOS/UEFI only with faculty or lab support.

Checkpoint

Host OS, RAM, free storage and virtualization status are recorded before downloading any installer.

Record 1 - Host readiness evidence



Stage 2 - Download the Selected Hypervisor

Learn: The hypervisor is the software layer that creates and manages the virtual computer.

- If choosing VirtualBox, open <https://www.virtualbox.org/wiki/Downloads> and select the platform package matching the host OS, such as Windows hosts or macOS hosts.
- If choosing VMware Workstation, use the faculty-shared installer or the official VMware/Broadcom Workstation download page and choose the installer matching the host OS.
- Save the installer inside the project folder or Downloads folder.
- Record installer filename, source link, download date and approximate file size.
- Do not download installers from unofficial software mirrors.

Checkpoint

The selected hypervisor installer file is available and its source is recorded.

Record 2 - Hypervisor download evidence

Search download history

Your browser is managed by your parent

Today

[VirtualBox-7.2.14-174565-Win \(1\).exe](https://www.virtualbox.org)

From <https://www.virtualbox.org>

[VirtualBox-7.2.14-174565-Win.exe](https://www.virtualbox.org)

From <https://www.virtualbox.org>

Download VirtualBox

The VirtualBox Extension Pack is available for personal and educational use on this page under the PUEL license. The VirtualBox Extension Pack is also available under commercial or enterprise terms. By downloading, you agree to the terms and conditions of the respective license.

VirtualBox Platform Packages

VirtualBox 7.2.14 platform packages

-  [Windows hosts](#)
-  [macOS / Intel hosts](#)
-  [macOS / Apple Silicon hosts](#)
-  [Linux distributions](#)
-  [Solaris hosts](#)
-  [Solaris 11 IPS hosts](#)

Platform packages are released under the terms of the [GPL version 3](#)

VirtualBox Extension Pack VirtualBox 7.2.14 Extension Pack

This VirtualBox Extension Pack Personal Use and Educational License governs your access to and use of the VirtualBox Extension Pack. It does not apply to the VirtualBox base package and/or its source code, which are licensed under version 3 of the GNU General Public License "GPL".

See our [FAQ](#) for answers to common questions.

VirtualBox Extension Pack Personal Use and Educational License (PUEL)

[PUEL License FAQ](#)

[PUEL License Text](#)

[Accept and download](#)

Oracle VirtualBox was downloaded from the official VirtualBox website.

Stage 3 - Install the Hypervisor

Learn: During installation, the hypervisor adds drivers for virtual networking, USB integration and virtual hardware management.

- For VirtualBox, open the installer, proceed with default components, allow network interface installation when prompted, finish installation and open VirtualBox Manager.
- For VMware Workstation, open the installer, choose Typical setup, keep default installation path, allow network adapter installation, skip or disable optional customer-experience participation if shown, finish and reboot if requested.
- For VMware Workstation personal/free use prompts, select the free/non-commercial option when it is legally available for the installed edition.
- Open the installed application and capture the home screen or About screen.
- Record the exact installed version.

Record 3 - Hypervisor installation evidence



Stage 4 - Choose and Download the Guest OS

Learn: The guest OS image controls download time, VM size and installation complexity.

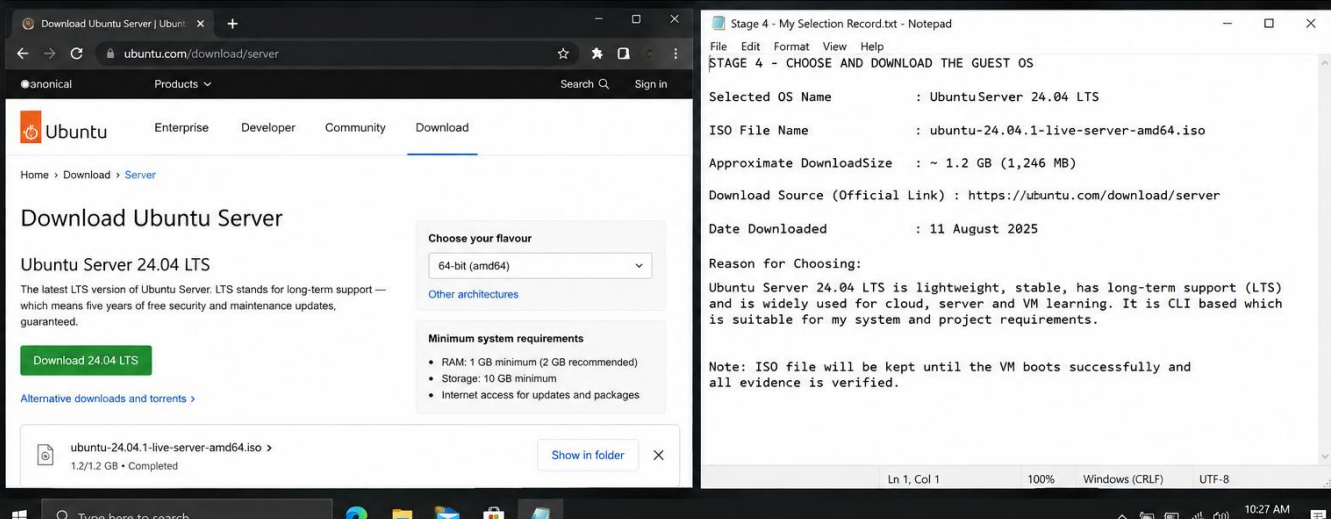
- Study the Guest OS Selection Guide table.
- Choose Tiny Core, Alpine, Debian netinst, Ubuntu Server or Ubuntu Desktop according to laptop capacity and internet speed.
- Download the ISO from the official link shown in the table.
- Record selected OS name, ISO filename, approximate download size and reason for choosing it. - Keep the ISO file until the VM boots successfully and evidence is verified.

Checkpoint

The selected guest OS ISO is ready and the selection reason is written clearly.

Record 4 - Guest OS selection evidence

Guest OS Selection Guide				
OS Option	Best For	Approx. ISO Size	Download Link (Official)	Notes
Tiny Core Linux	Very low-end systems, quick setup, minimal resources	~ 16 MB	https://tinycorelinux.net/	Extremely small, minimal OS, great for learning and basic tasks.
Alpine Linux	Low resource usage, security focused, container friendly	~ 130 MB	https://alpinelinux.org/downloads/	Lightweight, secure, easy to install.
Debian (netinst)	Stable, lightweight, good for servers and learning	~ 350 MB	https://www.debian.org/CD/netinst/	Network installer (netinst) is minimal and flexible.
Ubuntu Server	Server workloads, cloud, CLI based tasks	~ 1.2 GB	https://ubuntu.com/download/server	LTS support, reliable and widely used.
Ubuntu Desktop	GUI, ease of use, general purpose desktop experience	~ 3.0 GB	https://ubuntu.com/download/desktop	Full desktop environment, more resources required.



The screenshot shows the Ubuntu website's 'Download Ubuntu Server' page for version 24.04 LTS. It highlights the 'Download 24.04 LTS' button and the 'Choose your flavour' dropdown set to '64-bit (amd64)'. Below, the 'Minimum system requirements' are listed: RAM: 1 GB minimum (2 GB recommended), Storage: 10 GB minimum, and Internet access for updates and packages. The download progress bar shows 'ubuntu-24.04.1-live-server-amd64.iso' at 1.2/1.2 GB, completed.

The Notepad window titled 'Stage 4 - My Selection Record.txt' contains the following text:

```

File Edit Format View Help
STAGE 4 - CHOOSE AND DOWNLOAD THE GUEST OS

Selected OS Name       : UbuntuServer 24.04 LTS
ISO File Name          : ubuntu-24.04.1-live-server-amd64.iso
Approximate DownloadSize : ~ 1.2 GB (1,246 MB)
Download Source (Official Link) : https://ubuntu.com/download/server
Date Downloaded        : 11 August 2025

Reason for Choosing:
Ubuntu Server 24.04 LTS is lightweight, stable, has long-term support (LTS)
and is widely used for cloud, server and VM learning. It is CLI based which
is suitable for my system and project requirements.

Note: ISO file will be kept until the VM boots successfully and
all evidence is verified.
  
```

Stage 5 - Create the VM Shell

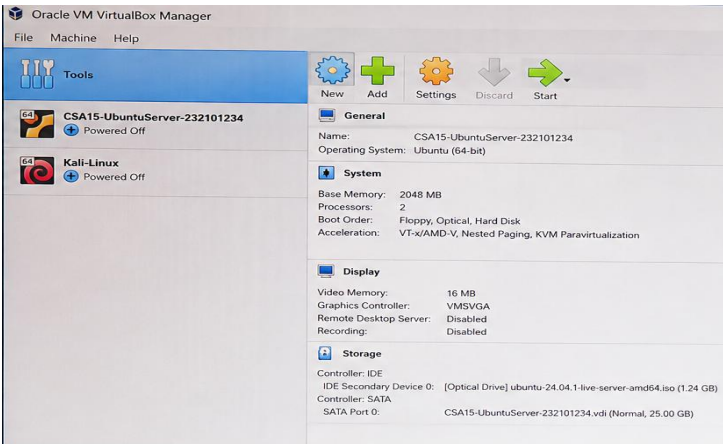
Learn: A VM shell is the empty virtual computer before the guest OS starts.

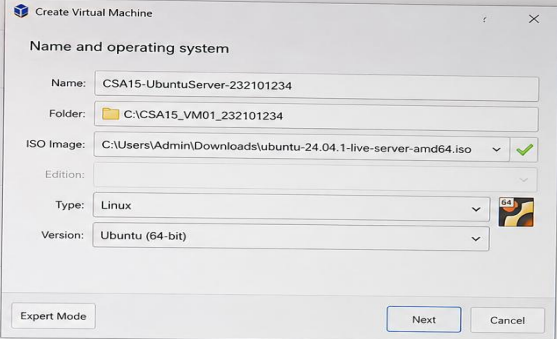
- In VirtualBox, click New. In VMware, choose Create a New Virtual Machine.
- Name the VM CSA15-SelectedOS-RegisterNumber.
- Choose Linux as the OS family and the closest matching version. Use Other Linux when the exact lightweight OS is not listed.
- Store the VM inside a known folder, preferably CSA15_VM01_RegisterNumber.
- Attach the selected ISO during creation if the wizard allows it; otherwise attach it later through VM settings.

Checkpoint

The new VM appears in VirtualBox Manager or VMware Library with the correct name and OS family.

Record 5 - VM creation evidence





STAGE 5 - CREATE THE VM SHELL

Learn: A VM shell is the empty virtual computer before the guest OS starts.

- In VirtualBox, click New. In VMware, choose Create a New Virtual Machine.
- Name the VM CSA15-SelectedOS-RegisterNumber.
- Choose Linux as the OS family and the closest matching version. Use Other Linux when the exact lightweight OS is not listed.
- Store the VM inside a known folder, preferably CSA15_VM01_RegisterNumber.
- Attach the selected ISO during creation if the wizard allows it; otherwise attach it later through VM settings.

CHECKPOINT: The new VM appears in VirtualBox Manager or VMware Library with the correct name and OS family.
Record 5 - VM creation evidence.

VM CREATION RECORD	
VM Name	CSA15-UbuntuServer-232101234
Folder Location	C:\CSA15_VM01_232101234
Operating System	Linux
Version	Ubuntu (64-bit)
ISO Attached	ubuntu-24.04.1-live-server-amd64.iso (1.24 GB)
Date Created	11 August 2025
Virtualization Platform	Oracle VM VirtualBox

Stage 6 - Configure CPU, RAM, Disk and Network

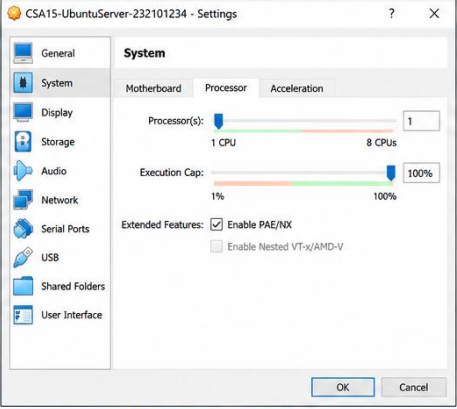
Learn: CPU, RAM and disk values must be enough for the guest OS without damaging host performance.

- Set CPU to 1 processor/core for the lab.
- Set RAM from the OS table. For the standard lab configuration use 2048 MB.
- Create a dynamically allocated disk. Use 15 GB for the standard lab configuration or the OS-table value for lightweight OS choices.
- Use VDI for VirtualBox or VMDK for VMware unless the faculty specifies another disk format.
- Keep the network adapter as NAT.
- Open VM settings and capture CPU, RAM, disk and network screens before first boot.

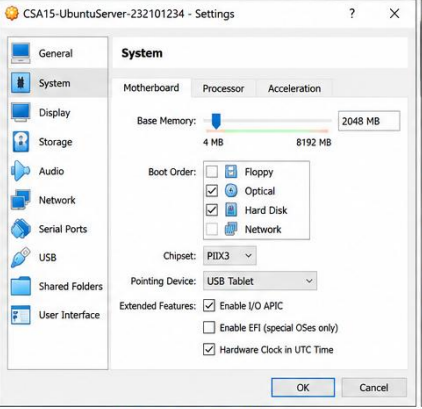
Checkpoint

These VM settings show CPU, RAM, virtual disk size, disk format/controller and NAT network.

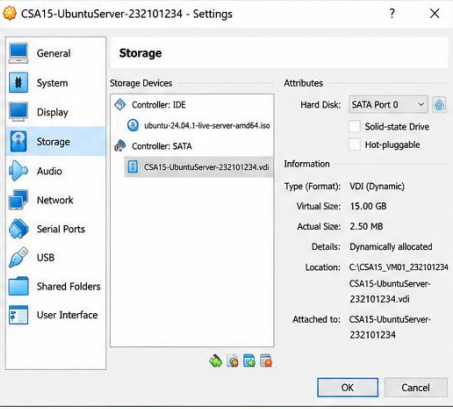
Record 6 - VM hardware configuration evidence



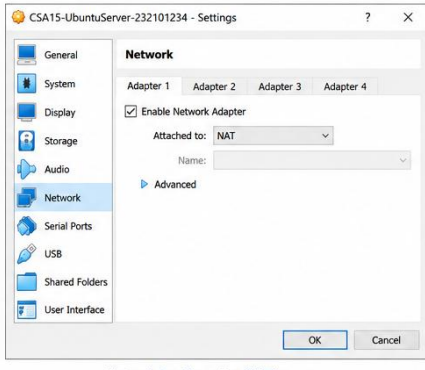
CPU Configuration (1 Processor/Core)



RAM Configuration (2048 MB)



Disk Configuration (15 GB; VDI, Dynamically Allocated)



Network Configuration (NAT)

RECORD 6 - VM HARDWARE CONFIGURATION EVIDENCE

Item	Configuration
VM Name	CSA15-UbuntuServer-232101234
CPU	1 Processor / 1 Core
RAM	2048 MB
Storage Controller	SATA
Virtual Disk	15.00 GB (Dynamically Allocated)
Disk Format	VDI (VirtualBox Disk Image)
ISO Attached	ubuntu-24.04.1-live-server-amd64.iso
Network Adapter	Adapter 1: NAT
Platform	Oracle VM VirtualBox

✓ **CHECKPOINT:** The VM settings show CPU, RAM, virtual disk size, disk format/controller and NAT network.



Stage 7 - Boot, Install or Run the Guest OS

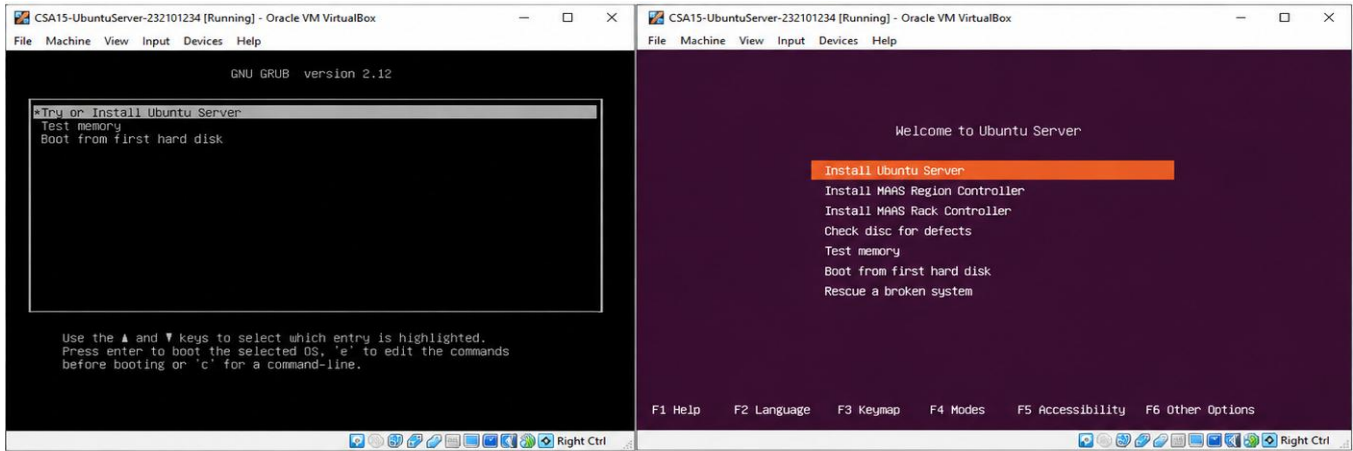
Learn: The VM boots from the ISO first; after installation, it can boot from the virtual disk.

- Start or Power On the VM.
- If the OS is Tiny Core, reach the live desktop and record the desktop proof.
- If the OS is Alpine or Debian, reach the installer or shell/login prompt and follow the minimal installation path if instructed.
- If the OS is Ubuntu, choose Try/Install or Install Ubuntu according to lab time and faculty instruction.
- When an installer asks about disk usage, choose the VM virtual disk only. It is safe to erase the VM disk; it must not refer to the host physical disk.
- After boot or installation, capture the guest OS screen.

Checkpoint

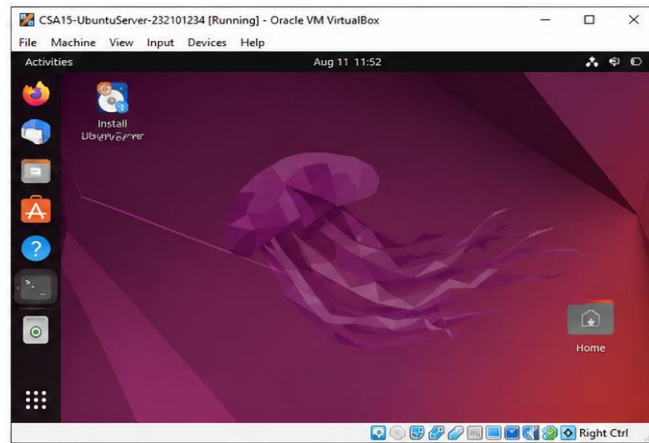
Three selected guest OS boots inside the VM and the evidence clearly shows it is running within the hypervisor window.

Record 7 - Guest OS boot or installation evidence



1. VM Start - Boot Menu (Ubuntu Server ISO)

2. Ubuntu Server Installer Menu



3. Ubuntu Server Live Desktop (Proof)

RECORD 7 - GUEST OS BOOT / INSTALLATION EVIDENCE

Item	Details
VM Name	CSA15-UbuntuServer-232101234
Selected OS	Ubuntu Server 24.04 LTS
Boot Source	ubuntu-24.04.1-live-server-amd64.iso (ISO)
Boot/Install Step Reached	Ubuntu Server Installer Menu / Live Desktop
Disk Selected for Installation	Virtual Disk (15.00 GB) - VDI (Dynamically Allocated)
Network Mode	NAT
Evidence Captured	Boot Menu, Installer Menu, Live Desktop
Date	11 August 2025



CHECKPOINT: The selected guest OS boots inside the VM and the evidence clearly shows it is running within the hypervisor window.

Stage 8 - Verify the VM from Inside the Guest OS

Learn: A configured VM is valid only when the guest OS can report its CPU, memory, storage and network state.

- Open terminal, shell or console in the guest OS.
- Run lscpu if available; otherwise record CPU details from the guest system or hypervisor settings.
- Run free -h or cat /proc/meminfo to verify memory.
- Run df -h to verify mounted storage.
- Run ip addr, ifconfig or ping 1.1.1.1 to verify networking.
- Capture all command outputs in a readable screenshot.

Checkpoint

CPU, RAM, disk and network evidence is captured from the guest OS or justified with available alternatives.

Record 8 - Guest verification evidence

```
CSA15-UbuntuServer-232101234 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Aug 11 12:15
ubuntu@ubuntu-server: ~
ubuntu@ubuntu-server:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                1
On-line CPU(s) list:   0
Vendor ID:             GenuineIntel
Model name:            Intel(R) Core(TM) i7-8565U CPU @ 1.80GHz
CPU family:            6
Model:                142
Thread(s) per core:    1
Core(s) per socket:    1
Socket(s):             1
Stepping:              11
BogoMIPS:              3599.99
Virtualization:        VT-x
Hypervisor vendor:     Oracle
Virtualization type:   full
L1d cache:             32K
L1i cache:             32K
L2 cache:              256K
L3 cache:              4096K
ubuntu@ubuntu-server:~$ free -h
               total        used        free      shared    buff/cache   available
Mem:            1.9Gi        340Mi       1.2Gi         9.0Mi        376Mi       1.5Gi
Swap:           975Mi         0B           0B
ubuntu@ubuntu-server:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           190M  1.1M  189M   1% /run
/dev/sda1       15G   1.4G   13G  10% /
tmpfs           947M   0 947M   0% /dev/shm
tmpfs           5.0M   0 5.0M   0% /run/lock
tmpfs           190M   0 190M   0% /run/user/1000
ubuntu@ubuntu-server:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3d:6a:1b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85594sec preferred_lft 85594sec
    inet6 fe80::a00:27ff:fe3d:6a1b/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
ubuntu@ubuntu-server:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (31.1.1.1) 56(84) bytes of data:
 64 bytes from 1.1.1.1: icmp_seq=1 ttl=58 time=11.5 ms
 64 bytes from 1.1.1.1: icmp_seq=2 ttl=58 time=11.6 ms
 64 bytes from 1.1.1.1: icmp_seq=3 ttl=58 time=11.5 ms
 64 bytes from 1.1.1.1: icmp_seq=4 ttl=58 time=11.5 ms
--- 1.1.1.1 ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 3011ms
 rtt min/avg/max/mdev = 11.527/11.560/11.615/0.036 ms
ubuntu@ubuntu-server:~$
```

STAGE 8 - VERIFY THE VM FROM INSIDE THE GUEST OS

Learn: A configured VM is valid only when the guest OS can report its CPU, memory, storage and network state.

- Open terminal, shell or console in the guest OS.
- Run `lscpu` if available; otherwise record CPU details from the guest system or hypervisor settings.
- Run `free -h` or `cat /proc/meminfo` to verify memory.
- Run `df -h` to verify mounted storage.
- Run `ip addr`, `ifconfig` or `ping 1.1.1.1` to verify networking.
- Capture all command outputs in a readable screenshot.



CHECKPOINT: CPU, RAM, disk and network evidence is captured from the guest OS or justified with available alternatives.

RECORD 8 - GUEST VERIFICATION EVIDENCE

Item	Details
VM Name	CSA15-UbuntuServer-232101234
OS	Ubuntu Server 24.04 LTS
CPU (lscpu)	1 CPU, Intel(R) Core(TM) i7-8565U @ 1.80GHz (Virtualized)
Memory (free -h)	1.9 GiB Total, 1.2 GiB Free (No Swap Used)
Storage (df -h)	/dev/sda1 15G, 13G Available, Mounted on /
Network (ip addr)	enp0s3 - 10.0.2.15/24 (NAT)
Connectivity (ping)	Ping to 1.1.1.1 - 4 packets transmitted, 4 received, 0% packet loss
Date & Time	11 August 2025, 12:15 PM
Evidence Captured	Terminal outputs (lscpu, free -h, df -h, ip addr, ping 1.1.1.1)

Final Evidence Checklist

Done	Checkpoint	Evidence to Capture
☐	Host readiness verified	Host OS, RAM, storage and virtualization status evidence.
☐	Hypervisor installed	VirtualBox or VMware installer/source/version and home screen evidence.
☐	OS choice justified	Selected OS, approximate ISO size, source link and reason for selection.
☐	VM configured	CPU, RAM, disk, disk format/controller and NAT network screenshots.
☐	Guest OS running	Boot/desktop/installer/login proof inside hypervisor window.
☐	Guest verification completed	CPU, RAM, disk and network command evidence.

Explain and Extend

Explain why VirtualBox or VMware is suitable as a type-2 hypervisor for this lab.

VirtualBox and VMware are suitable Type-2 hypervisors because they run on top of the host operating system and allow us to create and manage virtual machines without modifying the physical computer.

Reasons:

1. **Easy to install and use** – They provide a simple graphical interface for creating VMs.
2. **Runs on host OS** – They work on Windows, Linux, or macOS as applications.
3. **Supports multiple guest OSs** – Linux distributions such as Ubuntu, Debian, Alpine, and Tiny Core can be installed.
4. **Resource allocation** – CPU, RAM, disk, and network resources can be assigned to each VM.
5. **Safe for lab experiments** – The guest OS runs in an isolated virtual environment, protecting the host system.
6. **NAT networking** – The VM can access the internet through the host without requiring a separate physical network.
7. **Snapshots and recovery** – VM states can be saved and restored easily during experiments.

Conclusion:

VirtualBox or VMware is ideal for this lab because it provides a low-cost, flexible, and isolated environment for creating, configuring, running, and verifying guest operating systems on a normal laptop.

Explain why your selected OS is appropriate for your laptop and internet constraints.

Why the Selected OS is Appropriate

Selected OS: Ubuntu Server 24.04 LTS

Ubuntu Server is appropriate for my laptop and internet constraints because it requires fewer system resources than Ubuntu Desktop and has a relatively smaller ISO download size.

- It requires about **2 GB RAM** for comfortable VM usage.
- The ISO is approximately **1.2 GB**, so the download is manageable with a normal internet connection.
- It works well with the **1 CPU, 2048 MB RAM, and 15 GB virtual disk** configured for this lab.
- Ubuntu Server does not require a heavy graphical desktop, reducing the VM's resource usage.
- It provides a stable **LTS (Long-Term Support)** release suitable for cloud and virtualization experiments.
- It is widely used for **servers, cloud computing, and Linux-based lab activities**.

Conclusion: Ubuntu Server 24.04 LTS provides a good balance between download size, laptop resource usage, stability, and suitability for the Cloud Computing lab.

Explain whether your VM configuration matches the standard 1 CPU, 2 GB RAM and 15 GB disk requirement or why you used a justified lightweight alternative.

VM Configuration Justification

My VM configuration **matches the standard lab requirement**:

Resource	Standard Requirement	My Configuration
CPU	1 CPU/Core	1 CPU/Core
RAM	2 GB	2048 MB (2 GB)
Disk	15 GB	15 GB
Disk Type	VDI	VDI, Dynamically Allocated
Network	NAT	NAT

Explanation:

I used the standard configuration because Ubuntu Server 24.04 LTS can run comfortably with 1 CPU and 2 GB RAM for this lab. The 15 GB dynamically allocated virtual disk provides sufficient storage while using host storage only as needed. NAT networking allows the VM to access the internet through the host system.

Conclusion:

The VM configuration exactly matches the standard 1 CPU, 2 GB RAM, and 15 GB disk requirement, so no lightweight alternative was necessary.

Extend this setup to your capstone: what component could be tested in this VM?

Capstone Extension

This VM can be used to test the **backend/API component** of my capstone project.

For example, in **MediQueue AI – Hospital OP Queue and No-Show Prediction App**, the VM can be used to:

- Install Python and required libraries.
- Run the backend/API server.
- Test the AI no-show prediction module.
- Connect the application to a test database.
- Test API requests and responses between the application and backend.
- Verify basic network connectivity and server performance.

Conclusion:

The VM provides an isolated environment where the capstone backend and AI prediction component can be developed and tested safely before deploying it to the cloud.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM01-Type2-Hypervisor-VM-Creation.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance**Declaration**

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: T.PRAKEERTHI Register Number:192524279 Date:11/08/2026